

Virtual Walking Generator from Omnidirectional Video with Ground-dependent Foot Vibrations

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ABSTRACT

Virtual walking sensation can be experienced by combining optic flow and rhythmical foot vibrations. We hypothesized that the walking sensation would be enhanced by presenting foot vibrations matched with the ground in the video scene. Camera transitions were estimated from omnidirectional movies, and footstep patterns were generated. Then, vibrations matched to the ground in the scene were presented to users. We evaluated the system varying the vibration congruency with the grounds, and found that the ground-matched vibrations enhanced virtual walking sensation, but it depended on the ground type. The virtual walking sensation would contribute to improving virtual travel experiences.

Keywords: walking, tactile sensation, optic flow, omnidirectional video, virtual travel.

Index Terms: I.3.7 [Computer graphics]: Three-Dimensional Graphics and Realism—Virtual Reality; H.1.2 [Models and Principles]: User/Machine Systems—human factors

1 INTRODUCTION

Our travel experience is very limited during COVID-19 pandemic. Omnidirectional movies provide experiences of moving in distant places, and we can look around the distant scenes using a head-mounted display. We feel telepresence during observing the omnidirectional movies. However, it seems very different from travel experiences. We propose a system in which observers feel realistic walking sensations during observing any omnidirectional movies. The walking sensation would contribute to improving virtual travel experiences.

A sensation of walking can be elicited by a method with presenting optic flow to the eyes and vibrations to the feet simultaneously [1-2]. The vibrations to the forefeet and the heels of the left and right feet must be rhythmical as simulating walking footsteps [1]. However, in these studies, the vibrations were constant and independent of scenes or type of ground though the haptic sensation could be different depending on the ground. Humans can optimally integrate visual and haptic information [3]. Therefore, we hypothesized that the walking sensation would be enhanced by presenting foot vibrations matched with the ground in the video scene. We predicted that the walking sensation would be

better when the scene and the vibration were matched than they were unmatched or the only visual scene was presented.

2 METHODS

2.1 Participants

Fourteen naïve participants with normal eye acuity performed the experiment. They were provided informed consent before the experiment. The methods of the experiment were approved by the Ethical Committee for Human-Subject Research at the Toyohashi University of Technology.

2.2 Apparatus

A 360-degree camera (RICOH THETA Z1) was placed on a stabilizer (DJI RONIN SC) to take video images during walking. We used a computer and Unity (2018.4.28f1) to present monocular 360 degrees videos to a head-mounted display (HMD: HTC VIVE, 1080 [width] x 1200 [height] pixels, refresh rate 90Hz) and foot vibrations to four vibro-transducers (Acoupe Lab Vp408) placed each for the left and right forefeet and heels. To eliminate sounds from the vibro-transducers, a white noise (70 dBA) was presented to participants through a noise-canceling headphone.

2.3 Stimuli and Conditions

Four types of scenes in monocular 360-degree videos (3840 [width] x 1920 [height], 29.97fps) were prepared: Indoor corridor, Grass ground, Concrete paved road, Snow ground (Figure 1). All scenes were forward moving images at a roughly constant speed (1.26-1.41 m/s depending on scenes) for 30 s. The snow ground scene was taken from YouTube [4], and the others were taken by us.



Figure 1: Four types of grounds in the stimuli.

The foot vibrations for different scenes were manually chosen from the source asset of Unity (Floor and Ground of Classic Footstep SFX, Footstep (Snow and Grass)). The transition of the camera position and rotation information was estimated using OpenVSLAM [5]. From the estimated camera motions, left and right (2-channel) footstep timings were generated for each movie

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(81 cm/step, 1.56-1.74 steps/s depending on scenes). This timing generation was not real-time, but automatically calculated before the experiment, and adjusted manually to fit the scene. These 2-channel vibration timings were converted to 4 channels (the left and right forefeet and heels) by adding 105 ms difference between the heel and the forefoot.

There were three vibration-congruency conditions: the vibration was matched to the ground type (congruent condition), not matched (incongruent condition), or any vibration was not presented (no vibration condition). In the incongruent condition, one of three unmatched vibrations was randomly presented.

Participants performed 48 trials (4 ground types x 3 vibration congruency x 4 repetitions) in a random order.

2.4 Procedure

Participants were asked to answer a questionnaire after observing a stimulus for 30 s. The questionnaire consisted of items on walking-related sensations and items on perception of the ground.

Walking-related sensations were measured by Visual Analog Scale (VAS); The most left meant no sensation and the most right meant the same sensation as in the real environment [1].

W1. I felt as if my whole body was moving forward (self-motion).

W2. I felt as if I was walking forward (walking sensation).

W3. I felt as if my feet were striking the ground (leg action).

W4. I felt as if I was actually there in the scene (telepresence).

The ground perception was measured by VAS, where paired adjectives about ground properties were placed at the left and right ends.

G1. Smooth – Rough

G2. Soft – Hard

G3. Wet – Dry

G4. Slippery – Sticky

The data were digitalized to 0-100 after measurements. The questions were presented in a random order in each trial.

3 RESULTS

We conducted two-way repeated-measured ANOVAs for walking-related sensations (4 different grounds x 3 vibration congruency conditions; Figure 2).

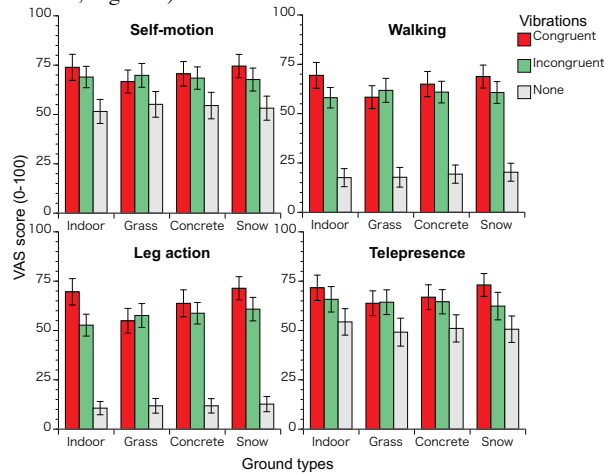


Figure 2: Results of walking-related sensations.

We found a significant main effect of the vibration congruency for all sensations [self-motion: $F(2, 26) = 19.52, p < .0001, \eta_p^2 = 0.60$, walking: $F(2, 26) = 54.05, p < .0001, \eta_p^2 = 0.81$, leg action: $F(2, 26) = 87.12, p < .0001, \eta_p^2 = 0.87$, telepresence: $F(2, 26) = 13.32, p < .0001, \eta_p^2 = 0.51$] and the interaction [self-motion: $F(6, 78) = 3.02, p = .010, \eta_p^2 = 0.19$, walking: $F(6, 78) = 2.55, p = .026, \eta_p^2 = 0.16$, leg action: $F(6, 78) = 4.04, p = .001, \eta_p^2 = 0.24$, telepresence: $F(6, 78) = 3.07, p = .010, \eta_p^2 = 0.19$]. For the leg

action and the telepresence, main effects of the grounds were significant [leg action: $F(3, 39) = 6.65, p = 0.001, \eta_p^2 = 0.34$, telepresence: $F(3, 39) = 5.10, p = 0.004, \eta_p^2 = 0.28$]. The analyses of simple main effects were performed ($\alpha = 0.05$). The self-motion sensation and the telepresence were higher in the congruent than the incongruent conditions, and higher in the incongruent than the no vibration condition in the snow ground scene, while they were higher in the congruent and incongruent conditions than the no vibration condition in the other ground scenes. The walking and leg-action sensations were higher in the congruent than the incongruent conditions, and higher in the incongruent than the no vibration condition in the inside corridor and snow ground scenes, while they were higher in the congruent and incongruent conditions than the no vibration condition in the other ground scenes.

We conducted two-way repeated-measured ANOVAs for ground-perception items (4 different grounds x 3 vibration congruency conditions), and found both main effects and the interaction for all items ($p < .01$). These results indicated that the different vibrations made the participants perceive different grounds, and the vibration congruency modulated it.

4 DISCUSSION AND CONCLUSION

The vibrations matched with the ground scenes improved all walking-related sensations over the incongruent vibration in the snow ground scenes, and improved the walking and leg-action sensations in the inside corridor scene. The foot vibrations improved all walking-related sensations over no vibration condition irrespective of congruency with the ground scene. These results suggest that the ground-matched vibration can contribute to enhancing virtual walking experience. However, it depends on the ground and vibration type. The tactile sensation during walking on the snow ground is very different from the other grounds, so that it would be suitable for our virtual walking system. The results also showed that the foot vibrations were able to induce appropriate perception of ground types. However, we should investigate how to optimize the vibration patterns for different grounds in a future.

We can make virtual walking experience from any omnidirectional movies with our proposed system; the system estimates the camera (walker's viewpoint) transition from the movies, provides appropriate footstep patterns, then presents appropriate foot vibrations to users. This system could improve our travel experience in virtual environments.

ACKNOWLEDGMENTS

This research was partly supported by JST ERATO Grant Number JPMJER1701 (Inami JIZAI Body Project), KAKENHI JP18H04118, and KAKENHI JP20H04489.

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